



**JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
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(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU & RTMNU



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**B. Tech Summer / Winter – 2026
(ESE / Make-up / Supplementary) Examination**

Branch: Computer Science & Engineering

Sem:- VI

Subject: PEC-V (Data Warehousing & Mining)

Subject Code: CS6E005A

Marks: 100

Date of Exam: 05/05/2026

Time: 3 Hrs.

Instructions to the students:

1. Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.
2. All the questions carry marks as indicated.
3. Assume suitable data whenever necessary.
4. Illustrate your answer whenever necessary with the help of neat sketches.
5. Use of Programmable Calculator is prohibited.
6. Don't use red pen for writing the answers.

COs: Students should be able to:

1. Understand the fundamental concepts, architecture, and components of data warehousing and business analysis.
2. Apply various data preprocessing and transformation techniques and understand the architecture and functionalities of data mining systems.
3. Analyse and implement classification and prediction techniques.
4. Evaluate and compare different clustering and outlier detection methods.
5. Explore and apply data mining techniques to complex data types.

SECTION A

1. All Questions are compulsory.
2. Answer in one/two word/sentence.

Q. No. Questions

Level/CO Mapped	Marks
L1/CO1	2
L1/CO3	2
L2/CO3	2
L1/CO4	2
L2/CO3	2
L2/CO2	2
L1/CO5	2
L3/CO4	2
L2/CO1	2
L1/CO3	2
L1/CO1	2
L1/CO2	2

- Q.1 Define Data Extraction.
- Q.2 State Decision Tree classification with a neat diagram.
- Q.3 Identify the different types of Ensemble learning.
- Q.4 List out and express the various Partitioning Clustering Method.
- Q.5 Rewrite about Support Vector Machines (SVM).
- Q.6 Classify the Data Mining Systems classified based on various factors.
- Q.7 State World Wide Web mining.
- Q.8 Write a short note on Grid-Based Clustering.
- Q.9 Draw the diagram for Components of Data warehouse.
- Q.10 Define Bayesian Classification.
- Q.11 State the term multidimensional data model.
- Q.12 Enlist the various data mining functionalities.

SECTION B

1. All Questions are compulsory.

Q. No.	Questions	Level/CO Mapped	Marks
Q.13	Distinguish between Data Warehouse and Data Mining.	L4/CO1	4
Q.14	Illustrate Metadata with various Types of Metadata in Data Warehousing.	L3/CO1	4
Q.15	Discuss the different types of Online Analytical Processing(OLAP) with their Advantages.	L2/CO1	4
Q.16	Examine the different steps for Building a data warehouse.	L4/CO1	4
Q.17	Write short note on Naïve Bayes Classifier.	L3/CO3	4
Q.18	Explain the various methods for evaluating the accuracy of a classifier or predictor.	L5/CO3	4

SECTION C

1. Solve any FOUR Questions.

Q. No.	Questions	Level/CO Mapped	Marks
Q.19	Illustrate the DBSCAN clustering method with different types of data points and also discuss its algorithm.	L4/CO4	7
Q.20	Explain the following: a) Types of data used in cluster analysis b) Types of Outlier Analysis.	L5/CO4	7
Q.21	Analyse the Three Tier Data Warehouse Architecture in detail	L4/CO1	7
Q.22	Examine the concept of OLAP and also Explain the following with Representation: a) Drill-Down b) Roll Up c) Slice d) Dice e) Pivot (Rotation)	L4/CO1	7
Q.23	Describe the different Hierarchical Clustering methods with Example.	L5/CO4	7
Q.24	Discuss the Architecture of a Typical Data Mining System with a neat diagram.	L5/CO2	7

SECTION D

1. Solve any TWO Questions.

Q. No.	Questions	Level/CO Mapped	Marks
Q.25	Rewrite about following in detail: a) Data Cleaning b) Data Integration c) Data Transformation d) Data Reduction	L5/CO2	12
Q.26	Explain DBMS Schemas for Decision Support including Star Schema, Snowflake Schema, and Fact Constellation with Example.	L6/CO1	12
Q.27	Describe Data Discretization and Concept Hierarchy Generation in detail with examples.	L6/CO2	12

Binary
ordinal
nominal